

Lecture 01 1. Compare and contrast the roles of an Intermediate System (IS) at the Data Link, Network, and Transport layers. Provide one practical example of each type. (4 marks)

An **Intermediate System (IS)** is a device that forwards data between end systems. Its role depends on the OSI/TCP-IP layer at which it operates.

a) Data Link Layer IS

- **Role:** Forwards frames based on **MAC addresses**.
- **Function:** Frame filtering, error detection, switching within the same LAN.
- **Example: Ethernet Switch**
 - A switch forwards Ethernet frames only to the correct port using MAC address tables.

b) Network Layer IS

- **Role:** Routes packets based on **logical (IP) addresses**.
- **Function:** Path selection, routing between different networks.
- **Example: Router**
 - A router forwards IP packets from a home network to the ISP network.

c) Transport Layer IS

- **Role:** Manages **end-to-end connections**, ports, and flow control.
- **Function:** Load balancing, connection termination, traffic inspection.
- **Example: Layer-4 Load Balancer**
 - Distributes TCP connections among multiple web servers using port numbers.

Comparison Summary

Layer	Address Used	Scope	Example
Data Link	MAC	Local network	Switch
Network	IP	Internetwork	Router
Transport	Port	End-to-end	Load balancer

2. Explain how the end-to-end principle and the “IP over everything” principle together form the foundation of the Internet’s design. Give a real-world example where these principles allow for easy adoption of new network technologies. (4 marks)

End-to-End Principle

- Intelligence is placed at the **end hosts**, not in the network core.
- The network core remains **simple**, while reliability and security are handled by endpoints.
- Examples: TCP reliability, encryption, error recovery.

“IP over Everything” Principle

- IP can run over **any physical or link-layer technology**.
- IP acts as a **universal abstraction layer**.
- Examples: IP over Ethernet, Wi-Fi, Fiber, Satellite, 5G.

How They Together Form Internet’s Foundation

- End-to-end principle ensures **application flexibility**.
- IP over everything ensures **technology independence**.
- Result: Easy innovation without changing the Internet core.

Real-World Example

- **Mobile Internet (4G/5G)**
 - New wireless technologies were deployed **without modifying TCP or HTTP**.
 - Smartphones run the same Internet applications as wired computers.

3. Analyze the differences between the OSI model and the TCP/IP model in terms of layer functionality. Specifically, discuss why the Presentation and Session layers are absent in the TCP/IP model and where their functions are implemented instead. (4 marks)

OSI vs TCP/IP: Missing Presentation & Session Layers (4 marks)

OSI Model (7 Layers)

1. Physical
2. Data Link
3. Network
4. Transport
5. **Session**
6. **Presentation**
7. Application

TCP/IP Model (4 Layers)

1. Network Interface
2. Internet
3. Transport
4. Application

Why Session & Presentation Layers Are Absent

- TCP/IP follows a **practical, implementation-driven design**.
- These layers were found unnecessary as separate layers in real networks.

Where Their Functions Exist Instead

OSI Layer **TCP/IP Implementation**

Session Application layer (HTTP sessions, TLS handshakes)

Presentation Application layer (encryption, compression, encoding)

Example

- HTTPS handles:
 - Encryption (Presentation)
 - Session management (Session)
 - All implemented inside the **Application layer**

5. Describe how protocol encapsulation and multiplexing/demultiplexing work together in the TCP/IP model. Use the example of an HTTP request being sent over Ethernet to illustrate the process. (4 marks)

Encapsulation & Multiplexing/Demultiplexing (4 marks)

Protocol Encapsulation

Each lower layer **wraps** the data with its own header before transmission.

Multiplexing

- Multiple applications share the same lower-layer protocol.
- Achieved using **port numbers**.

Demultiplexing

- Receiver uses headers to deliver data to the **correct application**.

Example: HTTP Request over Ethernet

Sender Side (Encapsulation):

1. HTTP request created (Application)
2. TCP adds port numbers (Transport)
3. IP adds source & destination IP (Internet)
4. Ethernet adds MAC addresses (Data Link)

Receiver Side (Demultiplexing):

1. Ethernet frame → IP packet
2. IP packet → TCP segment
3. TCP segment → HTTP application (port 80/443)

Result

Correct delivery of web data among many running applications.

5. Discuss the hierarchical structure of Internet network organization, from continental backbone networks down to individual hosts. Explain how this hierarchy supports scalability and interoperability across diverse network technologies.

Hierarchical Internet Structure (4 marks)

Internet Hierarchy Levels

1. **Continental / Global Backbone Networks**
 - High-capacity fiber links (Tier-1 ISPs)
2. **National / Regional ISPs**
 - Connect countries and regions
3. **Local ISPs**
 - Serve cities and towns
4. **Access Networks**
 - Home, campus, mobile networks
5. **End Hosts**
 - Computers, smartphones, servers

How Hierarchy Supports Scalability

- Routing tables remain manageable.
- Local changes don't affect global routing.
- Efficient traffic aggregation.

How It Ensures Interoperability

- Different technologies (fiber, Wi-Fi, cellular) interconnect using IP.
- Standardized protocols enable universal communication.

Example

- A smartphone in Bangladesh accesses a server in the USA seamlessly despite different access technologies.

Lecture 2 1. Explain how the IPv4 datagram header fields support fragmentation and reassembly across networks with different MTUs. Use a specific example involving Ethernet and X.25 networks to illustrate your explanation. (4 marks)

IPv4 supports fragmentation and reassembly to allow packets to travel across networks with different **Maximum Transmission Units (MTUs)**.

IPv4 Header Fields Used

- **Identification:** Same value in all fragments of a datagram; used for reassembly.
- **Flags:**
 - **DF (Don't Fragment):** Prevents fragmentation
 - **MF (More Fragments):** Indicates more fragments follow
- **Fragment Offset:** Indicates the position of the fragment in the original datagram.
- **Total Length:** Length of each fragment.

Example: Ethernet → X.25

- Ethernet MTU = **1500 bytes**
- X.25 MTU ≈ **576 bytes**
- A 1500-byte IP datagram from Ethernet must be **fragmented** when entering the X.25 network.
- The router splits the datagram into smaller fragments, each with:
 - Same **Identification**
 - Proper **Fragment Offset**
 - MF flag set except for the last fragment
- At the destination, fragments are **reassembled** using Identification and Fragment Offset.

Conclusion

These header fields ensure reliable delivery across heterogeneous networks.

2. Compare and contrast classful and classless addressing in IPv4. Discuss how CIDR alleviates the problem of address exhaustion and routing table explosion, providing an example with a given IP block. (4 marks)

Classful vs Classless Addressing and CIDR (4 marks)

Classful Addressing

- Fixed network/host boundaries
- Classes: A (/8), B (/16), C (/24)
- Inefficient use of IP addresses

- Large routing tables

Classless Addressing (CIDR)

- No fixed classes
- Uses **prefix length (/n)**
- Flexible allocation of IP addresses

How CIDR Solves Problems

1. **Address Exhaustion**
 - Allocates only required number of IPs
2. **Routing Table Explosion**
 - Supports **route aggregation (supernetting)**

Example

- Given IP block: **192.168.0.0/20**
- Represents **4096 IP addresses**
- Instead of advertising 16 separate /24 networks, a single **/20 route** is advertised.

Conclusion

CIDR improves scalability and efficiency of IPv4 addressing.

3. Describe the role of subnet masks in both subnetting and supernetting. Using the example of IP address 141.14.72.24 with subnet mask 255.255.192.0, show how the network and subnet addresses are derived and explain the impact on routing. (4 marks)

Role of Subnet Masks in Subnetting and Supernetting (4 marks)

Subnet Mask Purpose

- Separates **network** and **host** portions of an IP address
- Used in both:
 - **Subnetting**: Dividing a network
 - **Supernetting**: Combining networks

Given

- IP Address: **141.14.72.24**
- Subnet Mask: **255.255.192.0** (/18)

Binary Representation

- 255.255.192.0 →
11111111.11111111.11000000.00000000

Network Address Calculation

- Third octet block size = $256 - 192 = 64$
- Ranges: 0–63, **64–127**, 128–191
- 72 lies in **64–127**

→ Network Address: **141.14.64.0/18**

Impact on Routing

- Routers forward packets based on the **network prefix**
- Reduces routing table size via aggregation

4. Analyze the function of ICMPv4 in the IP network layer. Explain with examples how ICMP error-reporting messages (such as Destination Unreachable and Time Exceeded) assist in network diagnostics and troubleshooting.

Role of ICMPv4 in Network Diagnostics (4 marks)

Function of ICMPv4

- Supports **error reporting** and **network diagnostics**
- Works alongside IP (not a transport protocol)

Common ICMP Error Messages

1. **Destination Unreachable**
 - Indicates packet cannot reach destination
 - Causes: host down, port unreachable, network failure
 - Example: Server is offline → ICMP error sent to source
2. **Time Exceeded**
 - Generated when **TTL = 0**
 - Prevents infinite routing loops
 - Used by tools like **traceroute**

Diagnostic Role

- Helps identify:
 - Routing loops
 - Network congestion
 - Unreachable hosts or networks

Conclusion

ICMPv4 is essential for monitoring, troubleshooting, and maintaining IP network reliability.

Lecture 19 1. Explain why network addresses are considered "logical" and software-based, while MAC addresses are "physical" and hardware-based. Discuss how these two addressing layers interact in the process of Layer-2 and Layer-3 communication. (4 marks)

Logical (IP) vs Physical (MAC) Addresses and Their Interaction (4 marks)

Why IP Addresses Are Logical

- IP addresses are **software-configured**.
- Assigned dynamically (DHCP) or statically.
- Can change when a device moves to a different network.
- Used at the **Network Layer (Layer 3)** for routing.

Why MAC Addresses Are Physical

- MAC addresses are **hardware-based**, burned into the NIC by the manufacturer.
- Usually permanent and globally unique.
- Used at the **Data Link Layer (Layer 2)** for local delivery.

Interaction Between IP and MAC

- IP determines **where** the packet should go (logical destination).
- MAC determines **how** the packet is delivered on the local network.
- Before sending an IP packet, the sender resolves the destination MAC address using **ARP**.

Conclusion

Both addressing layers work together: IP for global routing, MAC for local frame delivery.

2. Describe the hierarchical nature of IP addressing and its significance in routing. How does this hierarchy help routers efficiently forward packets to destinations outside a host's subnet? (4 marks)

Hierarchical Nature of IP Addressing and Routing (4 marks)

Hierarchical Structure

IP addressing is divided into:

- **Network prefix** (network/subnet)
- **Host identifier**

Example:

192.168.10.25/24

- Network: 192.168.10.0
- Host: 25

Significance in Routing

- Routers forward packets based on the **network prefix**, not individual hosts.
- Enables **route aggregation (summarization)**.

Routing Outside the Subnet

- If destination IP is outside the local subnet:
 - Host forwards the packet to the **default gateway**.
 - Router uses longest-prefix match to select the best route.

Conclusion

Hierarchical IP addressing improves routing efficiency and Internet scalability.

3. Compare and contrast Unicast, Multicast, Broadcast, and Anycast addressing modes in terms of their purpose, packet delivery mechanism, and typical use cases in modern networks. (4 marks)

Addressing Modes: Unicast, Multicast, Broadcast, Anycast (4 marks)

Mode	Purpose	Delivery Mechanism	Typical Use Case
Unicast	One-to-one	One sender $\rightarrow$ one receiver	Web browsing, email
Broadcast	One-to-all	One sender $\rightarrow$ all hosts in subnet	ARP requests
Multicast	One-to-many	One sender $\rightarrow$ selected group	Video streaming, IPTV
Anycast	One-to-nearest	Delivered to closest node	DNS root servers

Key Differences

- Broadcast stays within a subnet.
- Multicast is group-based and bandwidth-efficient.
- Anycast improves performance and redundancy.

4.Explain the role of DNS in network addressing. Describe the process through which a host obtains the IP address of a remote host using DNS and how this address is used in subsequent packet forwarding. (4 marks)

Role of DNS in Network Addressing (4 marks)

Purpose of DNS

- Translates **human-readable domain names** into **IP addresses**.
- Eliminates the need to remember numeric IPs.

DNS Resolution Process

1. User enters a domain name (e.g., www.example.com).
2. Host checks local DNS cache.
3. If not found, query sent to DNS server.
4. DNS server returns the corresponding IP address.

Use in Packet Forwarding

- The obtained IP address is placed in the **destination IP field** of the IP packet.
- Routers forward packets based on this IP address.

Conclusion

DNS acts as the bridge between names and IP-based routing.

5.Analyze the routing process in a typical IP network. Discuss how routers use routing tables to determine the next hop, and why routers do not forward broadcast traffic by default. (4 marks)

Routing Process and Broadcast Handling (4 marks)

Routing Process

- Routers maintain a **routing table** containing:
 - Destination network
 - Next hop
 - Outgoing interface
- Routers use **longest prefix match** to choose the best route.

Next Hop Determination

- If destination is directly connected → send directly.
- Otherwise → forward to the next-hop router.

Why Routers Don't Forward Broadcasts

- Broadcast traffic is limited to a **broadcast domain**.
- Forwarding broadcasts would cause:
 - Network congestion
 - Broadcast storms

Conclusion

Blocking broadcast forwarding ensures network stability and scalability.

***Extra 1. Explain the process of Address Resolution Protocol (ARP) and its role in enabling communication between devices on the same broadcast domain. Discuss why ARP caching is important for network efficiency.**

Address Resolution Protocol (ARP) (Short Note)

ARP Process

1. Host sends ARP request (broadcast) asking for MAC of a known IP.
2. Destination replies with its MAC address.
3. Sender stores the mapping in the **ARP cache**.

Importance of ARP Caching

- Reduces repeated broadcasts.
- Improves network performance.
- Minimizes latency.

2. Describe the function of Internet Control Message Protocol (ICMP) in network diagnostics and error reporting. Explain how ICMP-echo and ICMP-echo-reply are used to test network connectivity and diagnose problems.

Internet Control Message Protocol (ICMP) (Short Note)

Function of ICMP

- Error reporting and network diagnostics.
- Operates at the Network Layer.

ICMP Echo & Echo Reply

- Used by **ping** command.
- Echo request sent → Echo reply received.
- Measures reachability, delay, and packet loss.

Diagnostic Value

- Detects unreachable hosts.
- Helps locate routing and connectivity problems.